

Reduced and Full classification of revivals on tori for separable coin in surd/fractional form

Ian Craig*

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1 Reduced Set of Revival Parameters

Observation 1 (Interchangeability of k and m). *The results are symmetric under the exchange $L_x \leftrightarrow L_y$: every parameter set $(\rho_x, \delta_x, \rho_y, \delta_y)$ that yields a revival of period N on $L_x \times L_y$ also yields a revival of the same period N on $L_y \times L_x$ with (ρ_x, δ_x) and (ρ_y, δ_y) interchanged. This follows from the symmetry of the separable coin $C = C_x \otimes C_y$ and the commutativity of the two shift operators. Consequently, only one orientation of each non-square torus needs to be tabulated; the $m \times k$ entries are obtained by swapping the x and y columns. The table below lists the reduced set.*

Table 1: Reduced exhaustive parameter sets yielding complete quantum revivals for separable quantum walks on $L_x \times L_y$ tori. Only one orientation of each non-square torus is listed; the $m \times k$ case follows by exchanging the x and y columns. Reflection parameters are exact nicefractions or surds; phase parameters are in units of π .

Torus	N	ρ_x	δ_x	ρ_y	δ_y
2×2	2	$1/2$	π	$1/2$	π
2×2	2	$1/2$	π	$1/2$	0
2×2	2	$1/2$	0	$1/2$	π
2×2	2	$1/2$	0	$1/2$	0
3×3	8	$2/3$	0	$2/3$	0
3×4	8	$2/3$	0	$1/2$	π
3×4	8	$2/3$	0	$1/2$	0
3×6	8	$2/3$	0	$2/3$	0
4×4	6	$3/4$	0	$3/4$	0

*Atlantic Technological University, Sligo, Ireland. ian.craig@atu.ie

Torus	T	ρ_x	δ_x	ρ_y	δ_y
4×4	8	$1/2$	π	$1/2$	π
4×4	8	$1/2$	π	$1/2$	0
4×4	8	$1/2$	0	$1/2$	π
4×4	8	$1/2$	0	$1/2$	0
4×6	8	$1/2$	π	$2/3$	0
4×6	8	$1/2$	0	$2/3$	0
6×6	8	$2/3$	0	$2/3$	0
3×3	10	$5-\sqrt{5}/6$	0	$5-\sqrt{5}/6$	0
3×6	10	$5-\sqrt{5}/6$	0	$5-\sqrt{5}/6$	0
6×6	10	$5-\sqrt{5}/6$	0	$5-\sqrt{5}/6$	0
3×3	12	$1/3$	$2\pi/3$	$1/3$	$4\pi/3$
3×3	12	$1/3$	$2\pi/3$	$1/3$	$2\pi/3$
3×3	12	$1/3$	$2\pi/3$	$1/3$	0
3×3	12	$1/3$	0	$1/3$	$2\pi/3$
3×3	12	$1/3$	0	$1/3$	0
3×3	12	$1/3$	$4\pi/3$	$1/3$	$2\pi/3$
3×3	12	$1/3$	$4\pi/3$	$1/3$	0
3×3	12	$1/3$	0	$1/3$	$4\pi/3$
3×3	12	$1/3$	$4\pi/3$	$1/3$	$4\pi/3$
3×4	12	$1/3$	0	$2-\sqrt{3}/2$	$\pi/2$
3×4	12	$1/3$	$4\pi/3$	$2-\sqrt{3}/2$	$3\pi/2$
3×4	12	$1/3$	$2\pi/3$	$3/4$	π
3×4	12	$1/3$	$4\pi/3$	$2-\sqrt{3}/2$	$\pi/2$
3×4	12	$1/3$	0	$3/4$	π
3×4	12	$1/3$	$2\pi/3$	$2-\sqrt{3}/2$	$3\pi/2$
3×4	12	$1/3$	$4\pi/3$	$1/4$	π
3×4	12	$1/3$	$2\pi/3$	$2-\sqrt{3}/2$	$\pi/2$
3×4	12	$1/3$	$4\pi/3$	$1/4$	0
3×4	12	$1/3$	0	$2-\sqrt{3}/2$	$3\pi/2$
3×4	12	$1/3$	$2\pi/3$	$1/4$	π
3×4	12	$1/3$	$2\pi/3$	$1/4$	0
3×4	12	$1/3$	0	$1/4$	π
3×4	12	$1/3$	0	$1/4$	0
3×4	12	$1/3$	$4\pi/3$	$3/4$	π
3×6	12	$1/3$	0	$1/3$	$4\pi/3$
3×6	12	$1/3$	0	$1/3$	$2\pi/3$
3×6	12	$1/3$	$4\pi/3$	$1/3$	$4\pi/3$
3×6	12	$1/3$	$4\pi/3$	$1/3$	$2\pi/3$
3×6	12	$1/3$	$2\pi/3$	$1/3$	$2\pi/3$
3×6	12	$1/3$	$2\pi/3$	$1/3$	0
3×6	12	$1/3$	0	$1/3$	0
3×6	12	$1/3$	$2\pi/3$	$1/3$	$4\pi/3$
3×6	12	$1/3$	$4\pi/3$	$1/3$	0
4×4	12	$2-\sqrt{3}/2$	$3\pi/2$	$1/4$	0

Torus	T	ρ_x	δ_x	ρ_y	δ_y
4×4	12	$2-\sqrt{3}/2$	$\pi/2$	$3/4$	π
4×4	12	$1/4$	0	$1/4$	π
4×4	12	$3/4$	π	$2-\sqrt{3}/2$	$\pi/2$
4×4	12	$1/4$	0	$1/4$	0
4×4	12	$1/4$	0	$2-\sqrt{3}/2$	$3\pi/2$
4×4	12	$1/4$	π	$3/4$	π
4×4	12	$2-\sqrt{3}/2$	$\pi/2$	$2-\sqrt{3}/2$	$3\pi/2$
4×4	12	$3/4$	π	$1/4$	π
4×4	12	$2-\sqrt{3}/2$	$\pi/2$	$2-\sqrt{3}/2$	$\pi/2$
4×4	12	$3/4$	π	$1/4$	0
4×4	12	$3/4$	π	$2-\sqrt{3}/2$	$3\pi/2$
4×4	12	$1/4$	π	$2-\sqrt{3}/2$	$3\pi/2$
4×4	12	$1/4$	π	$2-\sqrt{3}/2$	$\pi/2$
4×4	12	$2-\sqrt{3}/2$	$3\pi/2$	$3/4$	π
4×4	12	$2-\sqrt{3}/2$	$\pi/2$	$1/4$	0
4×4	12	$1/4$	π	$1/4$	π
4×4	12	$1/4$	π	$1/4$	0
4×4	12	$1/4$	0	$3/4$	π
4×4	12	$2-\sqrt{3}/2$	$3\pi/2$	$2-\sqrt{3}/2$	$3\pi/2$
4×4	12	$2-\sqrt{3}/2$	$3\pi/2$	$2-\sqrt{3}/2$	$\pi/2$
4×4	12	$2-\sqrt{3}/2$	$\pi/2$	$1/4$	π
4×4	12	$1/4$	0	$2-\sqrt{3}/2$	$\pi/2$
4×4	12	$2-\sqrt{3}/2$	$3\pi/2$	$1/4$	π
4×4	12	$3/4$	π	$3/4$	π
4×6	12	$1/4$	0	$1/3$	$2\pi/3$
4×6	12	$1/4$	0	$1/3$	0
4×6	12	$2-\sqrt{3}/2$	$3\pi/2$	$1/3$	$4\pi/3$
4×6	12	$2-\sqrt{3}/2$	$3\pi/2$	$1/3$	$2\pi/3$
4×6	12	$1/4$	0	$1/3$	$4\pi/3$
4×6	12	$3/4$	π	$1/3$	0
4×6	12	$1/4$	π	$1/3$	$4\pi/3$
4×6	12	$1/4$	π	$1/3$	$2\pi/3$
4×6	12	$1/4$	π	$1/3$	0
4×6	12	$2-\sqrt{3}/2$	$\pi/2$	$1/3$	$4\pi/3$
4×6	12	$2-\sqrt{3}/2$	$\pi/2$	$1/3$	$2\pi/3$
4×6	12	$3/4$	π	$1/3$	$4\pi/3$
4×6	12	$2-\sqrt{3}/2$	$\pi/2$	$1/3$	0
4×6	12	$3/4$	π	$1/3$	$2\pi/3$
4×6	12	$2-\sqrt{3}/2$	$3\pi/2$	$1/3$	0
6×6	12	$1/3$	$2\pi/3$	$1/3$	$4\pi/3$
6×6	12	$1/3$	$2\pi/3$	$1/3$	$2\pi/3$
6×6	12	$1/3$	$2\pi/3$	$1/3$	0
6×6	12	$1/3$	0	$1/3$	0

Torus	T	ρ_x	δ_x	ρ_y	δ_y
6×6	12	$1/3$	$4\pi/3$	$1/3$	0
6×6	12	$1/3$	0	$1/3$	$4\pi/3$
6×6	12	$1/3$	0	$1/3$	$2\pi/3$
6×6	12	$1/3$	$4\pi/3$	$1/3$	$4\pi/3$
6×6	12	$1/3$	$4\pi/3$	$1/3$	$2\pi/3$
3×3	16	$2-\sqrt{2}/3$	0	$2-\sqrt{2}/3$	0
3×4	16	$2-\sqrt{2}/3$	0	$2+\sqrt{2}/4$	0
3×4	16	$2-\sqrt{2}/3$	0	$2-\sqrt{2}/4$	0
3×4	16	$2-\sqrt{2}/3$	0	$2-\sqrt{2}/2$	$\pi/2$
3×6	16	$2-\sqrt{2}/3$	0	$2-\sqrt{2}/3$	0
4×4	16	$2-\sqrt{2}/4$	0	$2-\sqrt{2}/2$	$\pi/2$
4×4	16	$2+\sqrt{2}/4$	0	$2+\sqrt{2}/4$	0
4×4	16	$2-\sqrt{2}/2$	$\pi/2$	$2-\sqrt{2}/4$	0
4×4	16	$2-\sqrt{2}/4$	0	$2+\sqrt{2}/4$	0
4×4	16	$2-\sqrt{2}/2$	$\pi/2$	$2-\sqrt{2}/2$	$\pi/2$
4×4	16	$2+\sqrt{2}/4$	0	$2-\sqrt{2}/2$	$\pi/2$
4×4	16	$2+\sqrt{2}/4$	0	$2-\sqrt{2}/4$	0
4×4	16	$2-\sqrt{2}/2$	$\pi/2$	$2+\sqrt{2}/4$	0
4×4	16	$2-\sqrt{2}/4$	0	$2-\sqrt{2}/4$	0
4×6	16	$2-\sqrt{2}/2$	$\pi/2$	$2-\sqrt{2}/3$	0
4×6	16	$2-\sqrt{2}/4$	0	$2-\sqrt{2}/3$	0
4×6	16	$2+\sqrt{2}/4$	0	$2-\sqrt{2}/3$	0
6×6	16	$2-\sqrt{2}/3$	0	$2-\sqrt{2}/3$	0
3×3	20	$3-\sqrt{5}/6$	0	$3-\sqrt{5}/6$	0
3×4	20	$3-\sqrt{5}/6$	0	$3+\sqrt{5}/8$	0
3×4	20	$3-\sqrt{5}/6$	0	$5-\sqrt{5}/8$	π
3×4	20	$3-\sqrt{5}/6$	0	$3-\sqrt{5}/8$	0
4×4	20	$3+\sqrt{5}/8$	0	$3+\sqrt{5}/8$	0
4×4	20	$5-\sqrt{5}/8$	π	$3-\sqrt{5}/8$	0
4×4	20	$3-\sqrt{5}/8$	0	$5-\sqrt{5}/8$	π
4×4	20	$3-\sqrt{5}/8$	0	$3+\sqrt{5}/8$	0
4×4	20	$5-\sqrt{5}/8$	π	$3+\sqrt{5}/8$	0
4×4	20	$3+\sqrt{5}/8$	0	$3-\sqrt{5}/8$	0
4×4	20	$5-\sqrt{5}/8$	π	$5-\sqrt{5}/8$	π
4×4	20	$3+\sqrt{5}/8$	0	$5-\sqrt{5}/8$	π
4×4	20	$3-\sqrt{5}/8$	0	$3-\sqrt{5}/8$	0
3×3	24	$2/3$	$2\pi/3$	$2/3$	$2\pi/3$
3×3	24	$2/3$	$4\pi/3$	$2-\sqrt{3}/3$	0
3×3	24	$2/3$	$2\pi/3$	$2/3$	$4\pi/3$
3×3	24	$2/3$	$2\pi/3$	$2-\sqrt{3}/3$	0
3×3	24	$2-\sqrt{3}/3$	0	$2/3$	$4\pi/3$
3×3	24	$2-\sqrt{3}/3$	0	$2/3$	$2\pi/3$
3×3	24	$2/3$	$4\pi/3$	$2/3$	$4\pi/3$

Torus	T	ρ_x	δ_x	ρ_y	δ_y
3×3	24	$2/3$	$4\pi/3$	$2/3$	$2\pi/3$
3×3	24	$2-\sqrt{3}/3$	0	$2-\sqrt{3}/3$	0
3×4	24	$2/3$	$4\pi/3$	$2+\sqrt{3}/4$	0
3×4	24	$2/3$	$4\pi/3$	$1/2$	$\pi/2$
3×4	24	$2-\sqrt{3}/3$	0	$2+\sqrt{3}/4$	0
3×4	24	$2/3$	$4\pi/3$	$2-\sqrt{3}/4$	0
3×4	24	$2/3$	$2\pi/3$	$1/2$	$3\pi/2$
3×4	24	$2/3$	$2\pi/3$	$1/2$	$\pi/2$
3×4	24	$2/3$	$2\pi/3$	$2+\sqrt{3}/4$	0
3×4	24	$2-\sqrt{3}/3$	0	$2-\sqrt{3}/4$	0
3×4	24	$2-\sqrt{3}/3$	0	$1/2$	$3\pi/2$
3×4	24	$2-\sqrt{3}/3$	0	$1/2$	$\pi/2$
3×4	24	$2/3$	$2\pi/3$	$2-\sqrt{3}/4$	0
3×4	24	$2/3$	$4\pi/3$	$1/2$	$3\pi/2$
3×8	24	$2-\sqrt{3}/3$	0	$1/2$	$\pi/2$
3×8	24	$2/3$	$2\pi/3$	$1/2$	π
3×8	24	$2/3$	$4\pi/3$	$1/2$	$3\pi/2$
3×8	24	$2/3$	$4\pi/3$	$1/2$	$\pi/2$
3×8	24	$2-\sqrt{3}/3$	0	$1/2$	$3\pi/2$
3×8	24	$2-\sqrt{3}/3$	0	$1/2$	0
3×8	24	$2/3$	$2\pi/3$	$1/2$	$\pi/2$
3×8	24	$2/3$	$4\pi/3$	$1/2$	π
3×8	24	$2/3$	$4\pi/3$	$1/2$	0
3×8	24	$2-\sqrt{3}/3$	0	$1/2$	π
3×8	24	$2/3$	$2\pi/3$	$1/2$	$3\pi/2$
3×8	24	$2/3$	$2\pi/3$	$1/2$	0
4×4	24	$2-\sqrt{3}/4$	0	$1/2$	$3\pi/2$
4×4	24	$2+\sqrt{3}/4$	0	$1/2$	$3\pi/2$
4×4	24	$2+\sqrt{3}/4$	0	$1/2$	$\pi/2$
4×4	24	$2-\sqrt{3}/4$	0	$2+\sqrt{3}/4$	0
4×4	24	$1/2$	$3\pi/2$	$1/2$	$\pi/2$
4×4	24	$2+\sqrt{3}/4$	0	$2+\sqrt{3}/4$	0
4×4	24	$2-\sqrt{3}/4$	0	$2-\sqrt{3}/4$	0
4×4	24	$1/2$	$3\pi/2$	$1/2$	$3\pi/2$
4×4	24	$1/2$	$3\pi/2$	$2+\sqrt{3}/4$	0
4×4	24	$1/2$	$\pi/2$	$2-\sqrt{3}/4$	0
4×4	24	$2+\sqrt{3}/4$	0	$2-\sqrt{3}/4$	0
4×4	24	$1/2$	$3\pi/2$	$2-\sqrt{3}/4$	0
4×4	24	$2-\sqrt{3}/4$	0	$1/2$	$\pi/2$
4×4	24	$1/2$	$\pi/2$	$1/2$	$3\pi/2$
4×4	24	$1/2$	$\pi/2$	$1/2$	$\pi/2$
4×4	24	$1/2$	$\pi/2$	$2+\sqrt{3}/4$	0
4×8	24	$2+\sqrt{3}/4$	0	$1/2$	π

Torus	T	ρ_x	δ_x	ρ_y	δ_y
4×8	24	$1/2$	$\pi/2$	$1/2$	π
4×8	24	$1/2$	$\pi/2$	$1/2$	0
4×8	24	$1/2$	$3\pi/2$	$1/2$	$3\pi/2$
4×8	24	$1/2$	$3\pi/2$	$1/2$	$\pi/2$
4×8	24	$2+\sqrt{3}/4$	0	$1/2$	$\pi/2$
4×8	24	$2-\sqrt{3}/4$	0	$1/2$	$3\pi/2$
4×8	24	$2-\sqrt{3}/4$	0	$1/2$	$\pi/2$
4×8	24	$1/2$	$3\pi/2$	$1/2$	π
4×8	24	$1/2$	$3\pi/2$	$1/2$	0
4×8	24	$2+\sqrt{3}/4$	0	$1/2$	$3\pi/2$
4×8	24	$2+\sqrt{3}/4$	0	$1/2$	0
4×8	24	$1/2$	$\pi/2$	$1/2$	$3\pi/2$
4×8	24	$2-\sqrt{3}/4$	0	$1/2$	π
4×8	24	$1/2$	$\pi/2$	$1/2$	$\pi/2$
4×8	24	$2-\sqrt{3}/4$	0	$1/2$	0
8×8	24	$1/2$	$\pi/2$	$1/2$	π
8×8	24	$1/2$	0	$1/2$	$3\pi/2$
8×8	24	$1/2$	0	$1/2$	0
8×8	24	$1/2$	$3\pi/2$	$1/2$	π
8×8	24	$1/2$	$3\pi/2$	$1/2$	0
8×8	24	$1/2$	π	$1/2$	$3\pi/2$
8×8	24	$1/2$	π	$1/2$	$\pi/2$
8×8	24	$1/2$	0	$1/2$	π
8×8	24	$1/2$	$\pi/2$	$1/2$	$\pi/2$
8×8	24	$1/2$	π	$1/2$	π
8×8	24	$1/2$	π	$1/2$	0
8×8	24	$1/2$	$\pi/2$	$1/2$	$3\pi/2$
8×8	24	$1/2$	$\pi/2$	$1/2$	0
8×8	24	$1/2$	0	$1/2$	$\pi/2$
8×8	24	$1/2$	$3\pi/2$	$1/2$	$3\pi/2$
8×8	24	$1/2$	$3\pi/2$	$1/2$	$\pi/2$
3×3	30	$5-\sqrt{5}/6$	$2\pi/3$	$5-\sqrt{5}/6$	$4\pi/3$
3×3	30	$5-\sqrt{5}/6$	$4\pi/3$	$5-\sqrt{5}/6$	$4\pi/3$
3×3	30	$5-\sqrt{5}/6$	$4\pi/3$	$5-\sqrt{5}/6$	$2\pi/3$
3×3	30	$5-\sqrt{5}/6$	$2\pi/3$	$5-\sqrt{5}/6$	$2\pi/3$
5×5	60	$5-\sqrt{5}/10$	$4\pi/5$	$5-\sqrt{5}/10$	$4\pi/5$
5×5	60	$5+\sqrt{5}/10$	$4\pi/5$	$5-\sqrt{5}/10$	$2\pi/5$
5×5	60	$5+\sqrt{5}/10$	$2\pi/5$	$5+\sqrt{5}/10$	0
5×5	60	$5+\sqrt{5}/10$	0	$5-\sqrt{5}/10$	0
5×5	60	$5-\sqrt{5}/10$	$4\pi/5$	$5-\sqrt{5}/10$	0
5×5	60	$5+\sqrt{5}/10$	$2\pi/5$	$5-\sqrt{5}/10$	$4\pi/5$
5×5	60	$5+\sqrt{5}/10$	$2\pi/5$	$5+\sqrt{5}/10$	$4\pi/5$
5×5	60	$5-\sqrt{5}/10$	$2\pi/5$	$5+\sqrt{5}/10$	$2\pi/5$

Torus	T	ρ_x	δ_x	ρ_y	δ_y
5×5	60	$5+\sqrt{5}/10$	$4\pi/5$	$5+\sqrt{5}/10$	$4\pi/5$
5×5	60	$5-\sqrt{5}/10$	0	$5-\sqrt{5}/10$	$2\pi/5$
5×5	60	$5+\sqrt{5}/10$	$4\pi/5$	$5-\sqrt{5}/10$	$4\pi/5$
5×5	60	$5+\sqrt{5}/10$	$2\pi/5$	$5-\sqrt{5}/10$	0
5×5	60	$5+\sqrt{5}/10$	0	$5+\sqrt{5}/10$	$2\pi/5$
5×5	60	$5+\sqrt{5}/10$	0	$5-\sqrt{5}/10$	$2\pi/5$
5×5	60	$5-\sqrt{5}/10$	$4\pi/5$	$5+\sqrt{5}/10$	$2\pi/5$
5×5	60	$5+\sqrt{5}/10$	$4\pi/5$	$5+\sqrt{5}/10$	0
5×5	60	$5-\sqrt{5}/10$	$4\pi/5$	$5-\sqrt{5}/10$	$2\pi/5$
5×5	60	$5+\sqrt{5}/10$	$4\pi/5$	$5-\sqrt{5}/10$	0
5×5	60	$5-\sqrt{5}/10$	0	$5+\sqrt{5}/10$	$4\pi/5$
5×5	60	$5-\sqrt{5}/10$	0	$5-\sqrt{5}/10$	$4\pi/5$
5×5	60	$5-\sqrt{5}/10$	$2\pi/5$	$5-\sqrt{5}/10$	$2\pi/5$
5×5	60	$5+\sqrt{5}/10$	$2\pi/5$	$5-\sqrt{5}/10$	$2\pi/5$
5×5	60	$5+\sqrt{5}/10$	$2\pi/5$	$5+\sqrt{5}/10$	$2\pi/5$
5×5	60	$5-\sqrt{5}/10$	0	$5+\sqrt{5}/10$	0
5×5	60	$5+\sqrt{5}/10$	$4\pi/5$	$5+\sqrt{5}/10$	$2\pi/5$
5×5	60	$5-\sqrt{5}/10$	0	$5-\sqrt{5}/10$	0
5×5	60	$5+\sqrt{5}/10$	0	$5+\sqrt{5}/10$	0
5×5	60	$5-\sqrt{5}/10$	$4\pi/5$	$5+\sqrt{5}/10$	0
5×5	60	$5-\sqrt{5}/10$	$2\pi/5$	$5+\sqrt{5}/10$	$4\pi/5$
5×5	60	$5-\sqrt{5}/10$	$2\pi/5$	$5-\sqrt{5}/10$	$4\pi/5$
5×5	60	$5-\sqrt{5}/10$	0	$5+\sqrt{5}/10$	$2\pi/5$
5×5	60	$5-\sqrt{5}/10$	$2\pi/5$	$5+\sqrt{5}/10$	0
5×5	60	$5+\sqrt{5}/10$	0	$5+\sqrt{5}/10$	$4\pi/5$
5×5	60	$5-\sqrt{5}/10$	$2\pi/5$	$5-\sqrt{5}/10$	0
5×5	60	$5+\sqrt{5}/10$	0	$5-\sqrt{5}/10$	$4\pi/5$
5×5	60	$5-\sqrt{5}/10$	$4\pi/5$	$5+\sqrt{5}/10$	$4\pi/5$
5×10	60	$5-\sqrt{5}/10$	0	$5-\sqrt{5}/10$	0
5×10	60	$5-\sqrt{5}/10$	$4\pi/5$	$5+\sqrt{5}/10$	0
5×10	60	$5-\sqrt{5}/10$	0	$5+\sqrt{5}/10$	0
5×10	60	$5-\sqrt{5}/10$	$2\pi/5$	$5-\sqrt{5}/10$	0
5×10	60	$5+\sqrt{5}/10$	$2\pi/5$	$5+\sqrt{5}/10$	0
5×10	60	$5+\sqrt{5}/10$	$2\pi/5$	$5-\sqrt{5}/10$	0
5×10	60	$5+\sqrt{5}/10$	$4\pi/5$	$5+\sqrt{5}/10$	0
5×10	60	$5+\sqrt{5}/10$	0	$5+\sqrt{5}/10$	0
5×10	60	$5+\sqrt{5}/10$	0	$5-\sqrt{5}/10$	0
5×10	60	$5+\sqrt{5}/10$	$4\pi/5$	$5-\sqrt{5}/10$	0
5×10	60	$5-\sqrt{5}/10$	$2\pi/5$	$5+\sqrt{5}/10$	0
5×10	60	$5-\sqrt{5}/10$	$4\pi/5$	$5-\sqrt{5}/10$	0
10×10	60	$5-\sqrt{5}/10$	0	$5-\sqrt{5}/10$	0
10×10	60	$5-\sqrt{5}/10$	0	$5+\sqrt{5}/10$	0
10×10	60	$5+\sqrt{5}/10$	0	$5+\sqrt{5}/10$	0

Torus	T	ρ_x	δ_x	ρ_y	δ_y
10×10	60	$5+\sqrt{5}/10$	0	$5-\sqrt{5}/10$	0